Linear Regression Notes — English

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Atlas Skilltech University — ML Course Notes (Complete English)

1. Introduction to Linear Regression

What is Linear Regression?

Linear Regression is a **supervised learning algorithm** that models the relationship between input features and a **continuous target variable** by fitting a straight line (or hyperplane) through the data.

Core idea: Given historical data points, find the best-fit line that minimizes prediction error. Use that line to predict new, unseen values.

Real-world intuition: A hiring manager knows that salary typically increases with experience. If they plot salary vs. years of experience for 100 employees and draw the best-fit line, they can use it to estimate what to offer a new candidate with 7 years of experience. That is linear regression.

Types of Linear Regression

Type	Features	Use Case
Simple Linear Regression	1 input feature	Salary from experience
Multiple Linear Regression	2+ input features	House price from size, bedrooms, age
Polynomial Regression	Non-linear relationships	Curved data (uses polynomial features)

Mathematical Form

Simple Linear Regression:

```
y = B0 + B1*x + epsilon
```

Multiple Linear Regression:

```
y = B0 + B1*x1 + B2*x2 + ... + Bn*xn + epsilon
```

Matrix Form (used internally by sklearn):

```
y = X * beta + epsilon
beta = (X^T * X)^(-1) * X^T * y
```

- **y**: target variable (what we predict)
- **B0**: intercept (where the line crosses the y-axis)
- **B1...Bn**: coefficients (the weight/importance of each feature)
- **epsilon**: irreducible noise

Four Assumptions (LINE)

- **L**inearity: The relationship between X and Y is linear
- **I**ndependence: Observations are independent of each other
- **N**ormality: Residuals are normally distributed around zero
- **E**qual Variance (Homoscedasticity): The spread of residuals is constant across predicted values

Violating these assumptions does not necessarily break the model, but it does affect the reliability of confidence intervals and p-values.

Linear Regression vs Other Algorithms

Aspect	Linear Regression	Logistic Regression	Decision Tree
Output	Continuous	Probability (0-1)	Class label
Decision boundary	Hyperplane	S-curve	Step function
Interpretability	High	High	Medium
Handles non-linearity	Only with poly features	Only with poly features	Yes (natively)

2. Ordinary Least Squares (OLS)

What OLS Does

OLS finds the line (or hyperplane) that **minimizes the sum of squared residuals (SSR)**. It is the closed-form analytical solution to linear regression — no iteration needed, unlike gradient descent.

Why squared errors? Two reasons: (1) squaring makes all errors positive, so negative and positive errors do not cancel out; (2) squaring penalizes large errors more than small ones, pushing the line away from outliers.

OLS Formulas

Simple Linear Regression slope:

```
\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}
(x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}
```

Simple Linear Regression intercept:

```
\beta_0 = \bar{y} - \beta_1 \bar{x}
```

Matrix Form (Multiple Regression — the Normal Equations):

```
\hat{\beta} = (X^T X)^{-1} X^T y
```

OLS Derivation Intuition

To find the minimum of SSR, take the partial derivatives with respect to each parameter and set them to zero:

```
\frac{\partial \text{SSR}}{\partial \beta_0} = 0 \implies \beta_0 = \bar{y} -
\beta_1 \bar{x}
```

```
\frac{\partial \text{SSR}}{\partial \beta_1} = 0 \implies \beta_1 =
\frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}
```

This is why OLS is fast — it is a one-shot calculation with no iterations.

Python: OLS from Scratch

```
import numpy as np

# Simple Linear Regression
x_mean = np.mean(X)
y_mean = np.mean(y)

beta1 = np.sum((X - x_mean) * (y - y_mean)) / np.sum((X - x_mean)**2)
beta0 = y_mean - beta1 * x_mean

# Or using matrix form for multiple regression
X_b = np.c_[np.ones((len(X), 1)), X]  # add bias column
beta = np.linalg.inv(X_b.T @ X_b) @ X_b.T @ y
```

Goodness of Fit

Term	Formula	Meaning
SS_total	sum((y - y_mean)^2)	Total variance in y
SS_residual	sum((y - y_pred)^2)	Unexplained variance
SS_regression	SS_total - SS_res	Variance explained by the model
R-squared	1 - SS_res / SS_total	Fraction of variance explained

BLUE Property

Under the LINE assumptions, OLS is the **B**est **L**inear **U**nbiased **E**stimator:

- **Best**: minimum variance among all linear estimators
- **Unbiased**: E[beta_hat] = beta_true (no systematic error)
- This is the Gauss-Markov theorem

3. Cost Functions

What is a Cost Function?

A cost function (also called a **loss function**) quantifies how wrong the model is. The goal of training is to find the parameters that **minimize the cost function**.

Formal definition: A cost function maps the vector of parameters (B0, B1, ..., Bn) to a single scalar value representing the total prediction error across all training samples.

Mean Squared Error (MSE)

The standard cost function used by OLS and gradient descent for regression:

```
\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2
```

Properties:

- Penalizes large errors heavily (quadratic penalty)
- Differentiable everywhere — needed for gradient descent
- Sensitive to outliers
- Units are the square of the target variable units

Mean Absolute Error (MAE)

```
\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|
```

Properties:

- More robust to outliers than MSE
- Same units as the target variable
- Not differentiable at zero (requires subgradient methods)
- Preferred when the data has many outliers

Root Mean Squared Error (RMSE)

```
\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}
```

Properties:

- Same units as the target variable (unlike MSE)
- Most commonly reported regression metric
- Interpretable: "on average, predictions are off by X units"

R-Squared (Coefficient of Determination)

```
R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - \frac{\sum_i(y_i - \hat{y}_i)^2}{\sum_i(y_i - \bar{y})^2}
```

Interpretation:

- R^2 = 1.0: perfect prediction
- R^2 = 0.0: the model only predicts the mean (useless)
- R^2 < 0: the model is worse than predicting the mean
- R^2 = 0.72 means the model explains 72% of the variance in y

Important caveat: Adding more features always increases (or equals) R^2 on training data, even if they are noise. Use **Adjusted R^2** to penalize for extra features:

```
R^2_{\text{adj}} = 1 - (1 - R^2) \cdot \frac{n-1}{n-p-1}
```

where n = samples and p = number of features.

Choosing the Right Metric

Situation	Recommended Metric
Optimization and training	MSE (smooth gradient)
Outliers present in data	MAE
Reporting to stakeholders	RMSE (same units as target)
Comparing models on different datasets	R-squared
Many features, risk of overfitting	Adjusted R-squared

Python Code

```
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

y_pred = model.predict(X_test)

mse  = mean_squared_error(y_test, y_pred)
mae  = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2   = r2_score(y_test, y_pred)

# sklearn 1.4+ has built-in RMSE
rmse_direct = mean_squared_error(y_test, y_pred, squared=False)  # older API
# or:
from sklearn.metrics import root_mean_squared_error  # sklearn >= 1.4
```

4. Gradient Descent

What is Gradient Descent?

Gradient Descent is an **iterative first-order optimization algorithm** that finds the parameters minimizing a cost function by repeatedly moving in the direction of the steepest downhill slope.

Geometric interpretation: The cost function J(B0, B1) defines a surface in parameter space. Gradient descent rolls a ball down this surface toward the global minimum — at each step, the ball moves in the negative gradient direction by a distance proportional to the learning rate.

Update Rule

At each iteration, every parameter is updated simultaneously:

```
\beta_j := \beta_j - \alpha \frac{\partial J(\boldsymbol{\beta})}{\partial
\beta_j}
```

For linear regression with MSE cost, the closed-form gradients are:

```
\frac{\partial J}{\partial \beta_0} = -\frac{2}{n} \sum_{i=1}^{n}(y_i -
\hat{y}_i)
```

```
\frac{\partial J}{\partial \beta_j} = -\frac{2}{n} \sum_{i=1}^{n} x_{ij}(y_i - \hat{y}_i), \quad j \geq 1
```

Learning Rate Selection

The learning rate alpha controls the step size at each iteration:

Alpha	Behavior	Consequence
Too large (> 0.3)	Overshoots minimum	Cost increases, algorithm diverges
Too small (< 0.0001)	Tiny steps	Converges, but thousands of iterations needed
Optimal	Smooth decrease in cost	Fast, stable convergence

Practical starting points: Try 0.1, 0.01, 0.001 — plot the loss curve and pick the largest alpha that converges smoothly.

Three Variants

Variant	Data per Update	Characteristics	Use Case
Batch GD	Full training set	Stable, slow per epoch	Small datasets
Stochastic GD (SGD)	1 sample	Fast, noisy, may not converge	Online learning
Mini-Batch GD	32-256 samples	Balanced; standard in practice	Deep learning, large datasets

Implementation

```
import numpy as np

# Always normalize before GD (critical!)
X_norm = (X - X.mean()) / X.std()
y_norm = (y - y.mean()) / y.std()
```

```
n = len(X_norm)

beta0, beta1 = 0.0, 0.0
alpha = 0.01
epochs = 1000
loss_history = []

for epoch in range(epochs):
    y_pred = beta0 + beta1 * X_norm
    loss = np.mean((y_norm - y_pred)**2)
    loss_history.append(loss)

    grad_b0 = -(2/n) * np.sum(y_norm - y_pred)
    grad_b1 = -(2/n) * np.sum(X_norm * (y_norm - y_pred))

    beta0 -= alpha * grad_b0
    beta1 -= alpha * grad_b1

# sklearn's SGDRegressor uses stochastic GD
from sklearn.linear_model import SGDRegressor
sgd = SGDRegressor(alpha=0.0001, max_iter=1000, learning_rate='adaptive')
sgd.fit(X_train_scaled, y_train)
```

OLS vs Gradient Descent

Aspect	OLS (Closed Form)	Gradient Descent
Method	One matrix inversion	Iterative updates
Complexity	O(n * p^2)	O(n * p * iterations)
Best for	Small-to-medium datasets	Very large datasets
Memory	Needs full X^T * X	Mini-batch possible
Hyperparameters	None	Learning rate, epochs
sklearn class	`LinearRegression`	`SGDRegressor`

Rule of thumb: Use OLS (sklearn's `LinearRegression`) unless your dataset has millions of rows or you are building a neural network.

5. Sklearn Implementation + Residual Plots

Standard Pipeline

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.pipeline import Pipeline
import numpy as np

# 1. Split the data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 2. Build pipeline (scaler + model)
pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', LinearRegression())
])

# 3. Fit on raw training data (pipeline handles scaling)
pipeline.fit(X_train, y_train)

# 4. Predict on raw test data
y_pred = pipeline.predict(X_test)

# 5. Evaluate
print(f'R-squared:  {r2_score(y_test, y_pred):.4f}')
print(f'RMSE:       {np.sqrt(mean_squared_error(y_test, y_pred)):.2f}')
print(f'Intercept:  {pipeline.named_steps["model"].intercept_:.4f}')
print(f'Coefficients: {pipeline.named_steps["model"].coef_}')
```

Coefficient Interpretation

```
# Always use standardized coefficients for comparison
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
model_std = LinearRegression().fit(X_train_scaled, y_train)

coef_df = pd.DataFrame({
    'Feature': feature_names,
    'Std_Coef': model_std.coef_
}).sort_values('Std_Coef', key=abs, ascending=False)
```

Rule: A standardized coefficient of 0.5 means: when this feature increases by 1 standard deviation, the target increases by 0.5 standard deviations (holding all other features constant).

Residual Analysis — Method 1 (Classic)

Residual = y_actual - y_predicted. A well-fitted model has residuals that look like white noise.

```
import matplotlib.pyplot as plt
import seaborn as sns

residuals = y_test - y_pred

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# Residuals vs Predicted
axes[0].scatter(y_pred, residuals, alpha=0.5, s=20)
axes[0].axhline(0, color='red', linewidth=2, linestyle='--')
axes[0].set(xlabel='Predicted', ylabel='Residual',
            title='Residuals vs Predicted')

# Residual histogram
axes[1].hist(residuals, bins=30, edgecolor='white')
axes[1].axvline(0, color='red', linestyle='--')
axes[1].set(xlabel='Residual', title='Residual Distribution')

# seaborn residplot with LOWESS trend
sns.residplot(x=y_pred, y=y_test,
              lowess=True,
              scatter_kws={'alpha': 0.3},
```

```
              line_kws={'color': 'red'},
              ax=axes[2])
axes[2].set(title='Residual Plot + LOWESS Trend (seaborn)')

plt.tight_layout()
```

Residual Analysis — Method 2 (Modern sklearn)

```
from sklearn.metrics import PredictionErrorDisplay

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Actual vs Predicted
PredictionErrorDisplay.from_predictions(
    y_test, y_pred,
    kind='actual_vs_predicted',
    ax=axes[0]
)
axes[0].set_title('Actual vs Predicted (PredictionErrorDisplay)')

# Residuals vs Predicted
PredictionErrorDisplay.from_predictions(
    y_test, y_pred,
    kind='residual_vs_predicted',
    ax=axes[1]
)
axes[1].set_title('Residuals vs Predicted (PredictionErrorDisplay)')
```

Residual Pattern Diagnosis

Pattern Observed	What It Means	Fix
Random scatter around zero	Model assumptions met	No action needed
Funnel shape (variance increases)	Heteroscedasticity	Log-transform y, or weighted regression
Curved (U or inverted U)	Non-linearity not captured	Add polynomial or interaction features

Pattern Observed	What It Means	Fix
Distinct clusters	Missing categorical variable	Add the variable; check feature engineering
Outliers (high leverage points)	Extreme observations distorting fit	Robust regression (HuberRegressor)

6. Bias-Variance Tradeoff + Regularization

Bias and Variance

Every model's test error decomposes into three components:

```
\text{Expected Test MSE} = \text{Bias}^2 + \text{Variance} +
\sigma^2_{\text{noise}}
```

- **Bias**: Systematic error. How far off the model's average prediction is from the truth. Caused by overly simple models that miss real patterns (underfitting).
- **Variance**: How much the model's predictions change across different training sets. Caused by overly complex models that learn noise (overfitting).
- **Irreducible noise**: Variance inherent in the problem; cannot be eliminated.

The tradeoff: Reducing bias (more complex model) increases variance, and vice versa. The goal is to minimize their sum.

Model Complexity	Bias	Variance	Train Error	Test Error
Too simple (degree 1 for nonlinear data)	High	Low	High	High
Just right	Low	Low	Low	Low
Too complex (degree 10, few samples)	Low	High	Very Low	High

Regularization

Regularization adds a penalty on the magnitude of coefficients to the cost function. This constrains model complexity, reducing variance at the cost of a small increase in bias.

Key insight: Always scale features before applying regularization — otherwise the penalty is unfairly large for features with large scales.

Ridge Regression (L2 Penalty)

```
J_{\text{Ridge}}(\boldsymbol{\beta}) = \frac{1}{n}||\mathbf{y} -
X\boldsymbol{\beta}||^2 + \alpha ||\boldsymbol{\beta}||_2^2
```

- Shrinks all coefficients toward (but never to) zero
- Closed-form solution: beta = (X^T X + alpha I)^{-1} X^T y
- Best for: correlated features (multicollinearity), where many features contribute

Lasso Regression (L1 Penalty)

```
J_{\text{Lasso}}(\boldsymbol{\beta}) = \frac{1}{n}||\mathbf{y} -
X\boldsymbol{\beta}||^2 + \alpha ||\boldsymbol{\beta}||_1
```

- Can set coefficients to **exactly zero** (geometric property of the L1 ball's corners)
- Performs automatic feature selection
- Best for: sparse models where only a few features truly matter

ElasticNet (L1 + L2)

```
J_{\text{EN}} = \frac{1}{n}||y - X\beta||^2 + \alpha \left[ \rho ||\beta||_1
+ \frac{1-\rho}{2}||\beta||_2^2 \right]
```

- Combines the sparsity of Lasso with the stability of Ridge
- `l1_ratio` (rho): 0 = pure Ridge, 1 = pure Lasso

Python Implementation

```
from sklearn.linear_model import Ridge, Lasso, ElasticNet
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline

# Individual models
ridge = Ridge(alpha=1.0)
lasso = Lasso(alpha=0.1)
elastic = ElasticNet(alpha=0.1, l1_ratio=0.5)

# GridSearchCV for Ridge
param_grid = {'model__alpha': [0.001, 0.01, 0.1, 1, 10, 100]}
pipeline = Pipeline([('scaler', StandardScaler()), ('model', Ridge())])
grid = GridSearchCV(pipeline, param_grid, cv=5, scoring='r2')
```

```
grid.fit(X_train, y_train)  # pass raw data
print(f'Best alpha: {grid.best_params_}')
print(f'Best CV R²: {grid.best_score_:.4f}')

# Check Lasso sparsity
lasso_model = Lasso(alpha=0.1).fit(X_train_scaled, y_train)
zero_coefs = np.sum(lasso_model.coef_ == 0)
print(f'Lasso set {zero_coefs}/{X_train.shape[1]} features to zero')
```

Comparison

Property	Linear Reg	Ridge	Lasso	ElasticNet
Feature selection	No	No	**Yes**	**Partial**
Handles multicollinearity	No	**Yes**	Unstable	**Yes**
Sparsity	No	No	**Yes**	**Yes**
Hyperparameters	None	alpha	alpha	alpha, l1_ratio
Closed-form solution	Yes	Yes	No (coordinate descent)	No

Section 7: Feature Scaling, Cross Validation & Model Selection

7.1 Why Feature Scaling Matters

Consider a dataset with:

- Experience: 1 to 10 (years)
- Salary: 30,000 to 150,000 (USD)

Without scaling, salary would **dominate** simply due to its larger magnitude. Gradient Descent would converge poorly, and regularization penalties would be applied unevenly across features.

3 Core Scaling Methods:

Method	Formula	Output Range	Best For
StandardScaler	(x - mean) / std	~-3 to +3	Normally distributed data

Method	Formula	Output Range	Best For
MinMaxScaler	(x - min) / (max - min)	0 to 1	Bounded features, neural nets
RobustScaler	(x - Q2) / (Q3 - Q1)	Varies	Datasets with outliers

---

7.2 StandardScaler — Z-Score Normalization

```
z = \frac{x - \mu}{\sigma}
```

Where mu = feature mean, sigma = feature standard deviation. After scaling: mean = 0, standard deviation = 1.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)  # Fit on training data only
X_test_scaled = scaler.transform(X_test)         # Transform only - never fit
on test!
```

Critical Rule: Never call `fit_transform()` on test data. This causes **data leakage** — the model learns from test statistics, inflating performance metrics.

7.3 MinMaxScaler — Bounded [0, 1] Scaling

```
x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}
```

```
from sklearn.preprocessing import MinMaxScaler
mm_scaler = MinMaxScaler()
X_scaled = mm_scaler.fit_transform(X)
# All values are now in [0, 1]
```

Sensitive to outliers — a single extreme value compresses all others into a narrow range.

7.4 RobustScaler — Outlier-Resistant Scaling

```
x_{scaled} = \frac{x - Q_2}{Q_3 - Q_1}
```

Where Q1, Q2 (median), Q3 are quartiles. IQR = Q3 - Q1.

Robust because outliers have minimal effect on the median and IQR, unlike mean and standard deviation.

7.5 When to Scale

Always scale for:

- Gradient Descent optimization
- Ridge, Lasso, ElasticNet (regularization depends on coefficient magnitude)
- KNN, SVM, Neural Networks (distance/magnitude sensitive)

Scaling not required for:

- Tree-based models (Decision Tree, Random Forest, XGBoost)
- Features already on the same scale

7.6 Cross Validation — Robust Model Evaluation

Problem with Single Train-Test Split: One split gives one estimate — lucky or unlucky depending on which samples land where.

K-Fold Cross Validation Solution:

Divide data into K equal folds. Train K times:

- One fold = validation set
- Remaining K-1 folds = training set

Report mean +/- std of K scores.

```
from sklearn.model_selection import cross_val_score, KFold
from sklearn.linear_model import LinearRegression

model = LinearRegression()
kf = KFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(model, X, y, cv=kf, scoring='r2')

print(f"5-Fold R2 Scores: {scores.round(3)}")
print(f"Mean R2: {scores.mean():.3f} +/- {scores.std():.3f}")
```

7.7 Pipeline — Prevent Data Leakage

Best Practice: Always wrap your scaler and model in a Pipeline. The pipeline automatically applies `fit_transform` on training folds and `transform` on validation folds — eliminating any risk of leakage.

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge

pipeline = Pipeline([
    ('scaler', StandardScaler()),  # Step 1: Scale
    ('model', Ridge(alpha=1.0))    # Step 2: Model
])

cv_scores = cross_val_score(pipeline, X, y, cv=5, scoring='r2')
print(f"Pipeline CV R2: {cv_scores.mean():.3f} +/- {cv_scores.std():.3f}")
```

7.8 Model Comparison via Cross Validation

```
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet

models = {
    'Linear Regression': LinearRegression(),
    'Ridge (a=1)':       Ridge(alpha=1.0),
    'Lasso (a=0.1)':    Lasso(alpha=0.1),
    'ElasticNet':        ElasticNet(alpha=0.1, l1_ratio=0.5)
}

for name, model in models.items():
    pipe = Pipeline([('scaler', StandardScaler()), ('model', model)])
    scores = cross_val_score(pipe, X, y, cv=5, scoring='r2')
    print(f"{name:25s}: {scores.mean():.3f} +/- {scores.std():.3f}")
```

7.9 GridSearchCV — Automated Hyperparameter Tuning

```
from sklearn.model_selection import GridSearchCV

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', Ridge())
])

param_grid = {
    'model__alpha': [0.001, 0.01, 0.1, 1.0, 10.0, 100.0]
}

grid_search = GridSearchCV(pipeline, param_grid, cv=5, scoring='r2',
n_jobs=-1)
grid_search.fit(X_train, y_train)

print(f"Best Alpha: {grid_search.best_params_['model__alpha']}")
print(f"Best CV R2: {grid_search.best_score_:.3f}")
print(f"Test R2:    {grid_search.score(X_test, y_test):.3f}")
```

Note: `model__alpha` uses double underscore to reference a Pipeline step's parameter.

7.10 Section Summary

Concept	Key Takeaway	sklearn API
StandardScaler	Mean=0, Std=1	`StandardScaler()`
MinMaxScaler	Bounded [0,1]	`MinMaxScaler()`
RobustScaler	Outlier-robust	`RobustScaler()`
K-Fold CV	K evaluations, mean score	`KFold(n_splits=5)`
cross_val_score	Quick CV scores	`cross_val_score(model, X, y, cv=5)`
Pipeline	Prevents data leakage	`Pipeline([('scaler', ...), ('model', ...)])`
GridSearchCV	Automated tuning	`GridSearchCV(pipe, param_grid)`

Golden Rule: Always wrap scaling inside a Pipeline before cross-validation. Fitting a scaler on the full dataset before splitting is a subtle but critical form of data leakage.

Section 8: Real World Applications, Cheat Sheet & Capstone

8.1 Where Linear Regression Is Used

Industry	Problem	Key Features	Target
Real Estate	House Price Prediction	Size, Rooms, Location	Price
HR	Salary Estimation	Experience, Skills, City	Salary
Healthcare	Hospital Stay Duration	Age, Diagnosis, Tests	Days
Finance	Credit Score Modeling	Income, Debt, History	Score
Manufacturing	Quality Control	Temperature, Pressure	Defects
Marketing	Ad Revenue Forecasting	Budget, Platform, Season	Revenue

---

8.2 End-to-End Example — House Price Prediction

A production-style project walkthrough:

Step 1: Load & Explore Data

```
import pandas as pd
df = pd.read_csv('houses.csv')  # Size, Rooms, Age, Location, Price
print(df.head())
print(df.describe())
```

Step 2: EDA

```
import seaborn as sns
import matplotlib.pyplot as plt

# Correlation heatmap
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Feature Correlations')
plt.show()

# Target distribution
sns.histplot(df['Price'], bins=30, kde=True)
plt.title('House Price Distribution')
plt.show()
```

Step 3: Pipeline + GridSearch

```
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge

X = df.drop('Price', axis=1)
y = df['Price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', Ridge())
])

param_grid = {'model__alpha': [0.01, 0.1, 1, 10, 100]}
grid = GridSearchCV(pipeline, param_grid, cv=5, scoring='r2')
grid.fit(X_train, y_train)

print(f'Best Alpha: {grid.best_params_}')
print(f'Test R2: {grid.score(X_test, y_test):.4f}')
```

Step 4: Deploy

```
import joblib

# Save
joblib.dump(grid.best_estimator_, 'house_price_model.pkl')

# Load and predict
model = joblib.load('house_price_model.pkl')
new_house = [[1500, 3, 10, 2]]  # Size, Rooms, Age, encoded Location
predicted_price = model.predict(new_house)
print(f'Predicted Price: ${predicted_price[0]:,.0f}')
```

8.3 Coefficient Interpretation

```
import pandas as pd

coef_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.named_steps['model'].coef_
}).sort_values('Coefficient', key=abs, ascending=False)

print(coef_df)
```

Interpretation rules:

- **Positive coefficient** (+500): Feature increases target when it increases
- **Negative coefficient** (-300): Feature decreases target when it increases
- **Larger absolute value** = More influential feature
- **Standardized coefficients** allow direct comparison of feature importance (after StandardScaler)
- A coefficient of 0.5 on a standardized feature means: one standard deviation increase in that feature → 0.5 unit increase in target

---

8.4 Capstone Project — California Housing

Complete applied project using `fetch_california_housing`:

Dataset: 20,640 samples, 8 features, continuous target

Features:

- MedInc (Median household income in block)

- HouseAge (Median house age in block)
- AveRooms, AveBedrms, AveOccup (averages per household)
- Population (block population)
- Latitude, Longitude (geographic coordinates)

Target: MedHouseVal (Median house value, in $100,000 units)

```
from sklearn.datasets import fetch_california_housing

data = fetch_california_housing(as_frame=True)
df = data.frame

print(f"Shape: {df.shape}")
print(df.describe())
```

Key findings:

1. MedInc has the strongest positive correlation with price
2. Geographic features (Latitude/Longitude) matter: coastal areas are expensive
3. High AveOccup (crowding) negatively affects price
4. Prices are capped at $5.00 (data artifact — real values above this are clipped)

Capstone Parts:

- Part A: Feature Engineering (rooms-per-person, bedrooms-ratio)
- Part B: Regional Analysis (coastal vs inland price comparison)
- Part C: ElasticNet with GridSearchCV (tune both alpha and l1_ratio)
- Part D: Error Analysis (which samples have highest prediction errors and why)

8.5 Complete Cheat Sheet — Linear Regression

Algorithm Selection Guide:

Model	Use Case	sklearn Class	Key Hyperparams
LinearRegression	Baseline, interpretable	`LinearRegression()`	None
Ridge (L2)	Multicollinearity, all features matter	`Ridge(alpha=1)`	alpha

Model	Use Case	sklearn Class	Key Hyperparams
Lasso (L1)	Feature selection, sparse solution	`Lasso(alpha=0.1)`	alpha
ElasticNet (L1+L2)	Both regularizations needed	`ElasticNet(alpha, l1_ratio)`	alpha, l1_ratio
SGDRegressor	Very large datasets	`SGDRegressor()`	learning_rate

Metrics Quick Reference:

Metric	sklearn Function	Range	Lower/Higher is Better?
MSE	`mean_squared_error`	0 to inf	Lower
MAE	`mean_absolute_error`	0 to inf	Lower
RMSE	`sqrt(mean_squared_error)`	0 to inf	Lower
R2	`r2_score`	-inf to 1	Higher (closer to 1)

Production-Ready Workflow Template:

```
# ---- IMPORT ----
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.model_selection import train_test_split, cross_val_score,
GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
import numpy as np, joblib

# ---- SPLIT ----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ---- PIPELINE + GRIDSEARCH ----
pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('model', Ridge())
```

```
])

grid = GridSearchCV(
    pipe,
    {'model__alpha': [0.001, 0.01, 0.1, 1, 10, 100]},
    cv=5,
    scoring='r2',
    n_jobs=-1
)
grid.fit(X_train, y_train)

# ---- EVALUATE ----
y_pred = grid.predict(X_test)
print(f"Best alpha: {grid.best_params_}")
print(f"R2:   {r2_score(y_test, y_pred):.4f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.4f}")
print(f"MAE:  {mean_absolute_error(y_test, y_pred):.4f}")

# ---- SAVE ----
joblib.dump(grid.best_estimator_, 'linear_model.pkl')

# ---- LOAD & PREDICT ----
loaded_model = joblib.load('linear_model.pkl')
prediction = loaded_model.predict(X_new)  # No manual scaling needed!
```

8.6 Key Takeaways — What We Learned

Linear Regression covers:

1. **Introduction** — Supervised learning for continuous targets, LINE assumptions
2. **OLS** — Minimizing sum of squared residuals, closed-form beta formula, R-squared
3. **Cost Functions** — MSE, MAE, RMSE, R2: when to use each
4. **Gradient Descent** — Iterative optimization, learning rate, Batch/SGD/Mini-batch
5. **sklearn Implementation** — Complete pipeline, residual plots (old + new), joblib
6. **Bias-Variance** — Underfitting vs overfitting, Ridge/Lasso/ElasticNet regularization
7. **Scaling + CV** — StandardScaler, Pipeline, K-Fold CV, GridSearchCV
8. **Real World** — End-to-end project, coefficient interpretation, California Housing capstone

Foundations that carry forward to every algorithm:

- Features and targets
- Train-test-validation split
- Evaluation metrics (MSE, R2)
- Overfitting and regularization
- Cross-validation
- Pipeline and deployment with joblib

Linear Regression is the simplest and most interpretable model in supervised learning. Even when better models exist for a problem, it remains the essential **baseline** every project starts with.